

25PY101: Engineering Physics

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Summative preparation

QM, QBT

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Quantum dot

The energy band gap of bulk GaAs is 1.424 eV. Assume that we can make a 1D quantum dot from the bulk.

- (a) What should be the size of the 1D quantum dot to emit red photon?
- (b) Is it practically feasible to manufacture 1D quantum dot of GaAs that emits blue light.
[Hint: Count the number of atoms in the dot.]

Q2. Graphical / Sketch-Based Question

Sketch and label the energy band diagram of n-type semiconductor

1. at $T=0$ K, and
2. at $T=300$ K.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: